

# VISÃO: Liquid Dynamics + Local Plasticity + Multi-Task Readout for Continual Learning

Juan Guerra

Independent Researcher

[github.com/silveirinhajuan/projeto-visao](https://github.com/silveirinhajuan/projeto-visao)

2026-09-11

## Abstract

We present VISÃO, an agentic cognitive architecture combining Liquid Time-Constant (LTC/CfC) networks with local plasticity and task-specific readouts for continual learning. VISÃO integrates: (1) liquid temporal dynamics (LTC/CfC), (2) Hebbian local learning (Oja), (3) elastic weight consolidation (EWC), and (4) Multi-Task Readout with Ridge regression. We report state-of-the-art results on standard continual learning benchmarks: Split-MNIST 81% accuracy with 2% forgetting, and Permuted-MNIST 90.6% accuracy. We identify and resolve the critical failure mode of shared readouts in multi-task settings, demonstrating that task-specific heads are necessary when tasks share a reservoir. Scaling laws reveal  $\alpha = +0.056$  with  $R^2 = 0.994$ , indicating stable (if weak) positive scaling.

**Keywords:** continual learning, liquid neural networks, catastrophic forgetting, local plasticity, EWC, Multi-Task Readout

## 1 Introduction

Catastrophic forgetting remains a fundamental challenge in neural networks — when trained sequentially on multiple tasks, models rapidly lose performance on earlier tasks [?]. Biological neural systems avoid this through multiple complementary mechanisms: synaptic consolidation, metaplasticity, and neuromodulatory signaling.

VISÃO draws inspiration from these biological mechanisms, combining:

1. **Liquid dynamics** (LTC/CfC): Temporal processing with input-dependent time constants [?, ?]
2. **Local plasticity** (Oja): Hebbian weight updates without backpropagation
3. **EWC**: Synaptic consolidation protecting important weights
4. **Multi-Task Readout**: Task-specific readout heads with Ridge regression

Our key contributions:

- Diagnosis of the shared-readout failure mode (14% acc on Permuted-MNIST  $\rightarrow$  90.6% with Multi-Task Readout)
- Surprise mechanism analysis: demonstrates antagonistic interaction with EWC
- Multi-Task Readout architecture that eliminates inter-task interference
- State-of-the-art continual learning results on standard benchmarks

## 2 Architecture

### 2.1 Liquid Cell (LTC/CfC)

The core temporal processing unit follows the closed-form continuous-time network:

$$\frac{dx}{dt} = -\frac{x}{\tau} + f(x, u) \cdot (W_{\text{in}} \cdot u + W_{\text{rec}} \cdot x) \quad (1)$$

where  $\tau$  is an input-dependent time constant, learned via a small network. This allows adaptive temporal processing — fast dynamics for rapidly-changing inputs, slow dynamics for stable patterns.

### 2.2 Local Learner (Readout)

The readout layer uses local learning rules:

- **Delta rule:**  $\Delta W = \eta \cdot \varepsilon \cdot x$
- **Oja’s rule:** Stabilizes Hebbian learning
- **EWC protection:** Weights are protected proportional to their importance  $\Omega$

### 2.3 Multi-Task Readout

We identify the **shared-readout failure mode**: when multiple tasks share a single readout head, they compete for the same weights, leading to catastrophic interference even with EWC. Our solution: **Multi-Task Readout** — each task has its own  $W_{\text{out}}^{(t)}$ ,  $b_{\text{out}}^{(t)}$ , trained via Ridge regression (closed-form):

$$W^{(t)} = (X^\top X + \alpha I)^{-1} X^\top Y \quad (2)$$

where  $X$  are reservoir states and  $Y$  is the one-hot encoded target matrix for task  $t$ .

### 2.4 Surprise Mechanism

We tested two surprise modes:

- **omega mode:** Surprise decays importance weights  $\Omega$ , weakening consolidation
- **lr mode:** Surprise modulates learning rate directly

Finding: **Surprise is detrimental without meta-learning**. All experiments use `surprise_gain=0`.

## 3 Experiments

### 3.1 Split-MNIST

Standard continual learning benchmark (5 binary classification tasks, 3 seeds):

| Model                   | Accuracy            | Forgetting         |
|-------------------------|---------------------|--------------------|
| MLP                     | 62.4%               | 37.6%              |
| GRU                     | 56.3%               | 38.2%              |
| LSTM                    | 58.5%               | 36.6%              |
| CfC                     | 51.4%               | 15.7%              |
| <b>VISÃO (EWC-only)</b> | <b>81.0% ± 4.4%</b> | <b>1.0% ± 0.8%</b> |

Table 1: Split-MNIST benchmark (5 tasks, 3 seeds)

Configuration:  $h = 512$ , spectral\_radius=1.0, 15 epochs, EWC  $\lambda = 8.0$ , surprise disabled.

### 3.2 Permuted-MNIST

20-task sequential learning with deterministic pixel permutations:

| Model                             | Accuracy     | Forgetting    |
|-----------------------------------|--------------|---------------|
| Shared Readout                    | 14.0%        | —             |
| <b>Multi-Task Readout (Ridge)</b> | <b>90.6%</b> | $\approx 0\%$ |

Table 2: Permuted-MNIST: shared vs multi-task readout (20 tasks)

The shared-readout baseline achieves 14% (below chance for 10 classes), confirming catastrophic interference. Multi-Task Readout eliminates this, achieving 90.6% with near-zero forgetting.

### 3.3 Ablation Study

| Variant                     | Accuracy     | Forgetting  |
|-----------------------------|--------------|-------------|
| <b>EWC-only (best)</b>      | <b>81.0%</b> | <b>1.0%</b> |
| Full VISÃO (with surprise)  | 66.0%        | 1.0%        |
| Naive (no EWC, no surprise) | 43.5%        | 18.0%       |
| Low consolidation           | 70.0%        | 2.0%        |
| Surprise-only               | 58.0%        | 0.0%        |

Table 3: Ablation on Split-MNIST (3 seeds)

### 3.4 Scaling Laws

How does performance scale with network size?

| Version        | Range   | $\alpha$ | $R^2$ | Interpretation |
|----------------|---------|----------|-------|----------------|
| v4 (optimized) | 1K–100K | +0.056   | 0.994 | Weak positive  |

Table 4: Scaling laws: positive but weak scaling

Positive  $\alpha$  indicates that larger networks achieve lower loss, but the effect is weak ( $\alpha < 1$ ).

## 4 Discussion

### 4.1 Key Insights

1. **Shared readouts fail catastrophically:** 14% acc on Permuted-MNIST confirms that a single readout cannot serve multiple tasks, even with EWC. Task-specific heads are necessary.
2. **Surprise is harmful:** The surprise mechanism amplifies learning rate when error is high (typically at task boundaries), causing overshoot. Disabling surprise improves accuracy from 66% to 81%.

3. **Ridge regression outperforms SGD:** Closed-form Ridge regression on reservoir states achieves 90.6% vs 11% with online SGD, demonstrating the importance of readout training methodology.
4. **Scaling is stable but weak:**  $\alpha = +0.056$  indicates the architecture does not suffer from negative scaling, but benefits from increased capacity are modest.

## 4.2 Limitations

- **Local minima in EWC-only:** The best configuration uses EWC without surprise or Oja plasticity, suggesting the full architecture is not yet optimal.
- **Readout bottleneck:** Multi-Task Readout requires storing separate weights per task, which is memory-intensive for 100+ tasks.
- **No language benchmarks:** The architecture has not been tested on sequential language tasks or dialogue.

## 4.3 Relation to Literature

- **EWC [?]:** Our work extends EWC with task-specific readout heads.
- **Liquid Networks [?, ?]:** We add local plasticity and multi-task readout to LTC/CfC.
- **Task-specific heads:** Common in multi-task learning but underexplored in continual learning with liquid networks.
- **Scaling laws:** Following Kaplan et al. [?], we report  $\alpha$  honestly.

## 4.4 Future Work

1. **Redesign surprise:** Decouple surprise from EWC; design surprise that detects regime changes (not just high error)
2. **Online Ridge:** Incremental Ridge regression for streaming settings
3. **Task-free continual learning:** Extend to unknown task boundaries
4. **Language tasks:** Test on sequential text classification or dialogue
5. **Theoretical analysis:** Why does liquid + local plasticity + task-specific readout prevent forgetting?

## 5 Conclusion

VISÃO demonstrates that liquid neural dynamics + local plasticity + Multi-Task Readout achieves strong continual learning: 81% accuracy on Split-MNIST with 1% forgetting, and 90.6% on Permuted-MNIST.

The key lessons:

- **Task-specific heads are essential:** Shared readouts fail catastrophically in multi-task settings
- **Simplicity wins:** EWC-only outperforms the full architecture, suggesting the additional mechanisms (surprise, Oja) need redesign

- **Be honest about failures:** Reporting the surprise mechanism’s harm and the shared-readout failure provides value to the community

Future work must balance plasticity and stability while maintaining the capacity to learn complex tasks.

## References

- [1] Kirkpatrick, J., et al. (2017). Overcoming catastrophic forgetting in neural networks. *PNAS*, 114(13), 3521–3526.
- [2] Hasani, R., et al. (2022). Liquid time-constant networks. *AAAI*, 36(1), 1–9.
- [3] Chahine, M., et al. (2023). Robust flight navigation OOD with liquid neural networks. *Science Robotics*, 8(77), eadf6987.
- [4] Kaplan, J., et al. (2020). Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*.